

KUGGOREN, SwedenWMO: **023550**Lat: **61.703N**Lon: **17.522E**Elev: **33**StdP: **14.68**Time Zone: **1.00 (EUC)**Period: **96-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.4	12.0	-3.8	4.5	17.8	2.3	6.2	15.8	38.8	29.6	35.4	30.0	9.7	320	0.706

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	72.0	64.0	69.5	62.6	67.3	61.2	66.0	70.1	64.2	67.8	62.5	66.1	9.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	90.3	68.5	62.6	85.1	66.6	60.9	79.9	64.8	30.7	70.2	29.4	68.0	28.2	66.2	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.5	27.3	23.5	DB	3.0	77.8	6.0	3.4	-1.3	80.3	-4.8	82.2	-8.2	84.2	-12.5	86.6
			WB	1.6	66.9	5.4	2.4	-2.2	68.7	-5.4	70.1	-8.4	71.4	-12.3	73.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.9	28.4	28.5	32.4	38.2	46.1	54.4	61.3	60.4	53.3	43.7	36.2	31.2	
	DBStd	13.10	7.75	8.21	7.20	5.46	5.19	4.59	4.21	4.12	4.75	5.15	5.87	7.37	
	HDD50	3547	669	601	547	355	144	11	0	0	21	202	415	582	
	HDD65	8079	1134	1021	1012	803	587	318	131	151	350	660	865	1047	
	CDD50	971	0	0	0	2	23	144	351	324	120	7	0	0	
	CDD65	28	0	0	0	0	0	1	17	10	0	0	0	0	
	CDH74	35	0	0	0	0	0	2	24	9	0	0	0	0	
	CDH80	4	0	0	0	0	0	0	3	1	0	0	0	0	
Wind	WSAvg	10.9	12.5	12.2	11.1	10.1	9.9	10.0	9.4	9.6	10.2	11.5	12.0	12.2	
Precipitation	PrecAvg	18.0	1.0	0.9	0.7	1.1	1.5	1.6	2.0	2.7	1.5	2.0	1.7	1.4	
	PrecMax	27.3	2.7	1.4	1.9	2.8	2.6	2.8	4.8	5.5	3.0	4.8	4.3	3.2	
	PrecMin	12.2	0.1	0.4	0.0	0.2	0.2	0.0	0.6	0.4	0.5	0.0	0.5	0.2	
	PrecStd	3.5	0.7	0.3	0.4	0.7	0.7	0.9	1.2	1.3	0.7	1.4	1.1	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.9	46.9	52.6	59.1	64.9	71.0	76.6	74.5	69.5	58.5	52.2	48.2	
		MCWB	38.8	40.1	43.0	48.2	50.6	58.4	64.9	64.0	59.3	52.6	47.0	43.5	
	2%	DB	41.6	44.2	48.4	53.0	59.5	66.5	73.0	71.3	64.6	54.7	48.1	44.6	
		MCWB	37.0	38.5	40.3	43.9	50.3	57.1	64.9	63.8	57.2	50.9	44.4	41.0	
	5%	DB	39.6	41.5	45.4	49.0	56.6	63.7	70.4	69.0	62.3	52.8	45.3	42.3	
		MCWB	36.0	37.0	38.9	41.9	49.4	55.8	64.0	62.7	56.6	49.7	43.0	39.3	
	10%	DB	37.6	38.4	42.4	45.8	53.7	61.4	68.0	66.9	60.1	50.8	43.6	40.4	
		MCWB	35.0	34.7	36.9	40.2	47.5	54.7	62.2	61.6	55.5	47.7	41.9	37.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.4	41.3	44.2	48.4	54.4	61.1	69.1	67.9	61.4	54.3	48.1	44.9
			MCDB	43.3	45.8	51.5	57.4	60.3	67.2	72.8	71.9	65.6	56.2	51.5	47.2
2%		WB	38.3	39.2	41.4	44.9	51.8	58.8	66.9	65.7	59.3	52.2	45.3	41.9	
		MCDB	40.5	43.3	47.2	51.1	57.8	64.3	71.1	69.4	62.9	54.2	47.4	43.9	
5%		WB	36.8	37.3	39.3	42.6	49.9	57.2	64.9	64.3	57.8	50.3	43.6	40.0	
		MCDB	38.7	41.1	44.7	47.7	55.4	62.0	69.0	67.4	60.7	52.6	45.0	41.8	
10%		WB	35.4	35.4	37.1	40.8	48.1	55.8	63.0	62.8	56.6	47.9	42.3	38.1	
		MCDB	37.1	37.9	41.7	45.1	52.9	60.1	67.0	66.0	59.2	50.0	43.4	40.0	
Mean Daily Temperature Range	5% DB	MDBR	7.3	7.8	8.5	8.7	8.7	8.3	8.2	8.4	8.4	7.0	6.0	7.0	
		MCDBR	8.3	9.8	12.9	15.4	14.4	13.3	11.9	10.8	11.8	9.4	7.8	8.4	
		MCWBR	6.8	7.1	8.0	8.8	7.7	7.1	6.5	6.3	7.2	7.0	6.8	7.0	
	5% WB	MCDBR	7.2	9.5	12.4	14.3	13.0	11.4	9.6	8.4	9.4	8.3	7.1	7.5	
		MCWBR	6.4	7.4	8.3	8.7	7.8	7.0	6.3	5.9	6.6	7.0	6.6	6.8	
Clear-Sky Solar Irradiance	taub	0.253	0.304	0.324	0.359	0.360	0.354	0.374	0.367	0.334	0.320	0.271	0.343		
	taud	2.169	2.283	2.373	2.372	2.437	2.494	2.458	2.471	2.544	2.437	2.208	2.580		
	Ebn at Noon	146	211	252	264	273	277	267	259	250	206	134	94		
	Edh at Noon	11	20	26	31	32	30	31	28	22	17	10	7		
All-Sky Solar Radiation	RadAvg	72	272	752	1222	1561	1771	1652	1255	796	358	104	42		
	RadStd	10	50	85	104	163	118	170	110	100	61	22	4		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A